

Homework#2 (Part I)

CS576 Machine Learning, Fall 2025

HW2 consists of **two parts (Part I and Part II)**.

- This document covers **Part I: Conceptual/Theoretical Understanding**.
- See the separate file for **Part II: Hands-On Practice**.

Part I. Conceptual/Theoretical Understanding

Due Date: October 12

Submission & Packing

- Submit a single file named YourLastName_YourFirstName_CS576_HW2_PartI.pdf (or .docx)
- Clearly label all answers within the file. Example: Q1 (1), Q1 (2), etc.

Grading Policy (Completion-Based)

- Each question is graded on completion.
- To receive full credit, provide a **substantive answer** for every sub-question.
- Missing any required item in a question will result in 0 points for that question.

Q1. Error types

Answer briefly:

- (1) Define **training error**, **validation error**, **test error**, and **true (generalization) error**.
- (2) Which one (training, validation, or test error) is directly related to the concept of **generalization error**, and why?

Q2. Error Estimation Methods

Compare the following methods of estimating error:

- Holdout sampling
- k-fold cross-validation
- Leave-one-out cross-validation
- Bootstrapping

For each method, describe **(1)** how it works, **(2)** its advantages, and **(3)** limitations.

Q3. Exploration Question: Bias–Variance Tradeoff

- (1) Define **bias** and **variance** in terms of model prediction errors.
- (2) Describe how high bias and high variance affect **training error** and **test error**, and explain the connection to underfitting and overfitting.
- (3) Explain the **bias–variance tradeoff curve**: why error is high at both extremes (overly simple vs. overly complex models), and why the best generalization occurs at an intermediate complexity.

Q4. Performance Metrics for Classification

Suppose you are given a model's predictions for a categorical target on a test set.

ID	Target	Prediction
1	FALSE	FALSE
2	FALSE	TRUE
3	FALSE	FALSE
4	FALSE	FALSE
5	TRUE	TRUE
6	FALSE	FALSE
7	TRUE	TRUE
8	TRUE	FALSE
9	FALSE	FALSE
10	FALSE	FALSE

ID	Target	Prediction
11	FALSE	FALSE
12	TRUE	TRUE
13	FALSE	TRUE
14	TRUE	TRUE
15	FALSE	FALSE
16	FALSE	FALSE
17	TRUE	FALSE
18	TRUE	TRUE
19	TRUE	TRUE
20	TRUE	TRUE

Provide:

- (1) The **confusion matrix**
- (2) The **misclassification rate**
- (3) The **average class accuracy** (use the **harmonic mean**)
- (4) The **precision, recall, and F1 measure**

Q5. Distance Metrics in k -NN

Answer the following:

- (1) Explain why the choice of distance metric matters in k -NN.
- (2) Compare **Euclidean**, **Manhattan**, and **Cosine similarity**, and describe the types of data where each metric is most appropriate.
- (2) Provide an example of a dataset or scenario where Euclidean distance might fail but cosine similarity works better. Clearly justify why cosine similarity is more suitable in this case.
- (3) Explain why **feature scaling** matters in k -NN.

Q6. Exploration Question: Effect of k in k -NN

Answer the following:

- (1) Explain how the choice of k influences the **bias–variance tradeoff** in k -NN classification.
- (2) As k increases, describe how the **decision boundary** changes. Illustrate with a sketch or description, and explain how this relates to **model complexity**.

Q6. Advanced Topics in k -NN

Answer the following:

- (1) Describe how predictions are made in **k-NN regression**
- (2) Define the “**curse of dimensionality**” in the context of *k*-NN
- (3) Explain why **irrelevant or redundant features** can hurt *k*-NN performance more than tree-based learners.

Q7. Bayes and Naïve Bayes Concepts

Answer the following:

- (1) State **Bayes’ theorem** and explain its importance in machine learning.
- (2) Explain **conditional independence** in probabilistic models. Provide an example in **text classification**.
- (3) Derive the **decision rule** for the Naïve Bayes classifier.
- (4) Explain the need for **Laplace (additive) smoothing** in Naïve Bayes

Q8. The table below lists a dataset of policyholders at an insurance company. The descriptive features include: Policyholder ID, Occupation, Gender, Age, Policy Type (Type of insurance policy), and Preferred contact channel (*target feature*).

ID	Occupation	Gender	Age	Policy Type	Pref Channel
1	lab tech	female	43	planC	email
2	farmhand	female	57	planA	phone
3	biophysicist	male	21	planA	email
4	sheriff	female	47	planB	phone
5	painter	male	55	planC	phone
6	manager	male	19	planA	email
7	geologist	male	49	planC	phone
8	messenger	male	51	planB	email
9	nurse	female	18	planC	phone

Answer the following:

- (1) Using **equal-frequency binning**, transform the **Age** feature into a categorical feature with three levels: **young**, **middle-aged**, and **mature**.
- (2) Examine the descriptive features in the dataset and identify which features should be **excluded** before building a predictive model. For each excluded feature, explain your reasoning.
- (3) Calculate the **probabilities required by a naive Bayes model** to represent this domain.
- (4) Based on these probabilities, determine which **target level** a naive Bayes model would predict for the following query:

Gender = female, Age = 30, Policy = planA